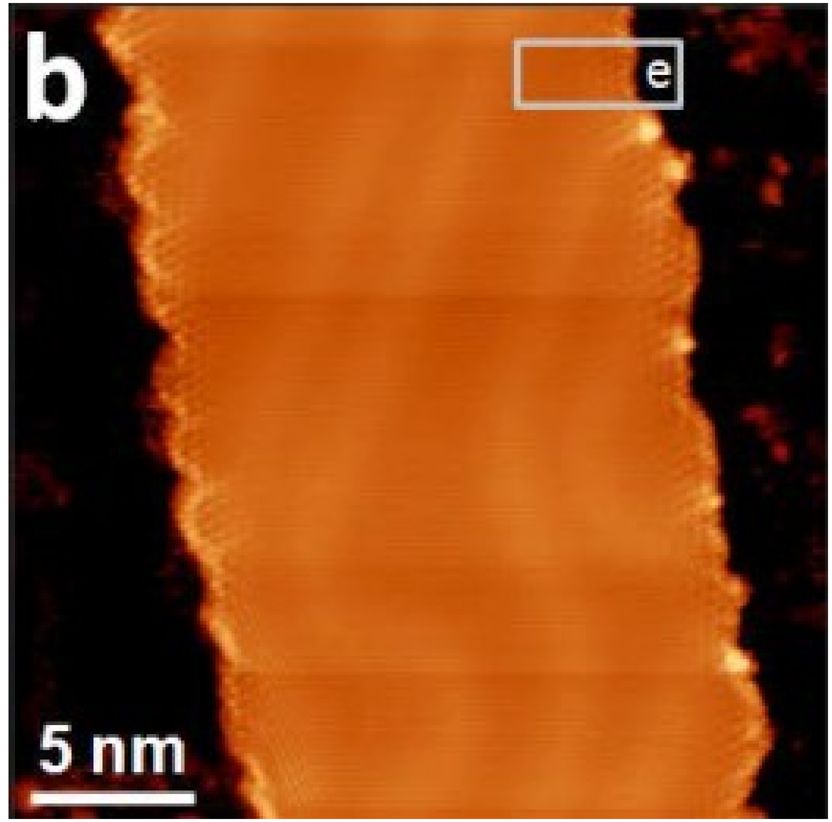
Magnetic Structure of Chiral Graphene Ribbons

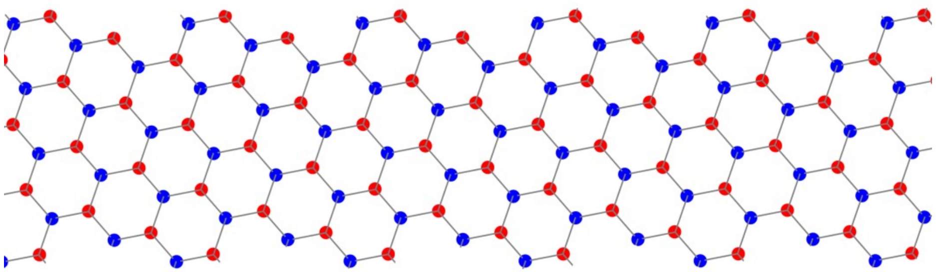
Kevin Pierce

Introduction

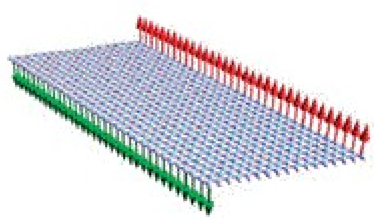
Different edge geometries possible



Carbon forms hexagonal ribbons



Simplest geometries
show magnetic edges

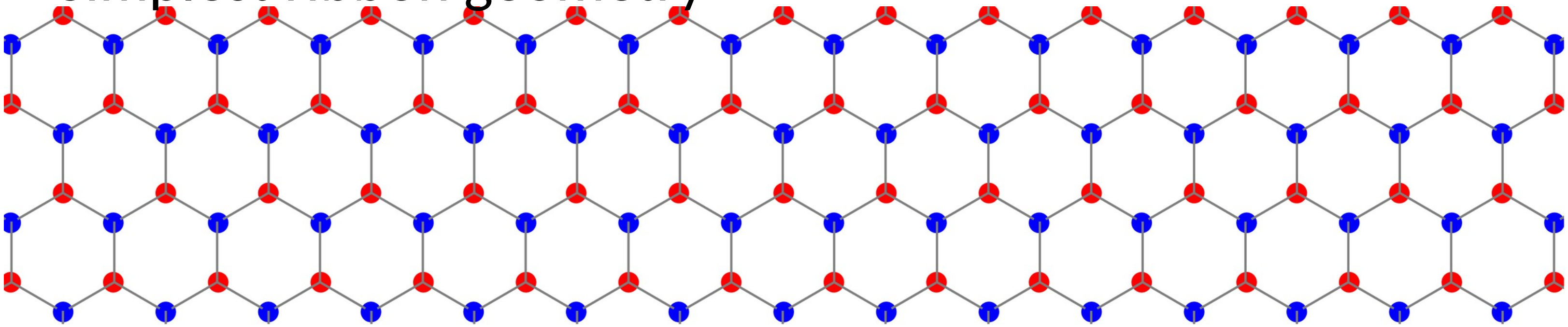


Need to understand

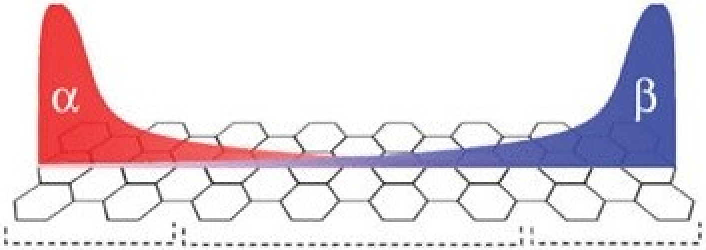
Can this be manipulated? Information? Spin-tronics?

Zigzag geometry

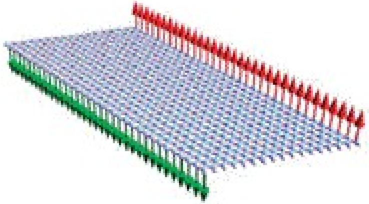
Simplest ribbon geometry



Low energy electrons localize to the edges

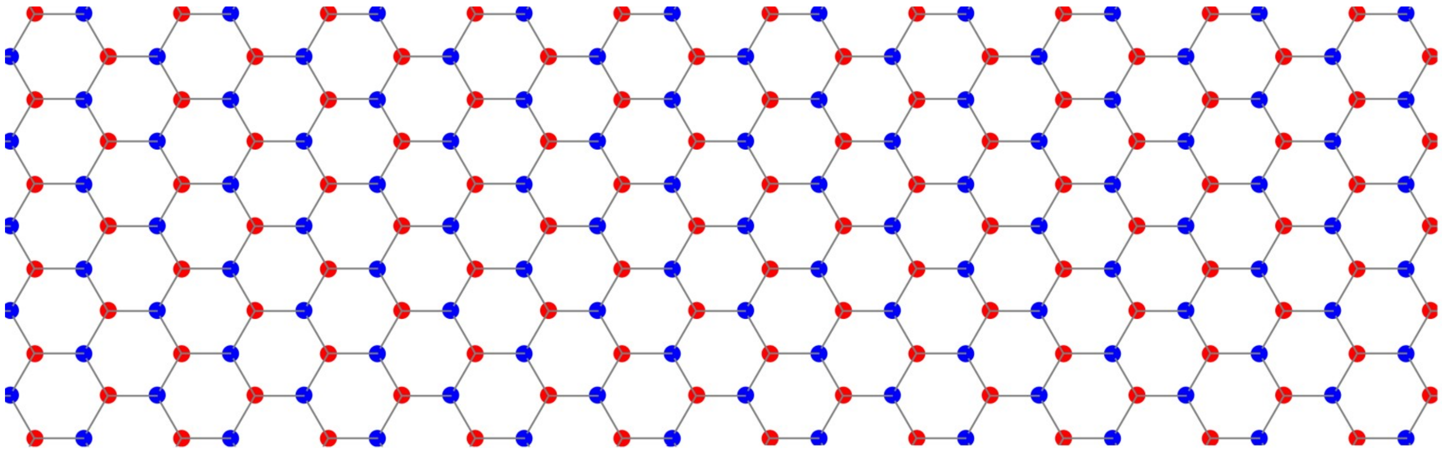


Edge electrons magnetically order



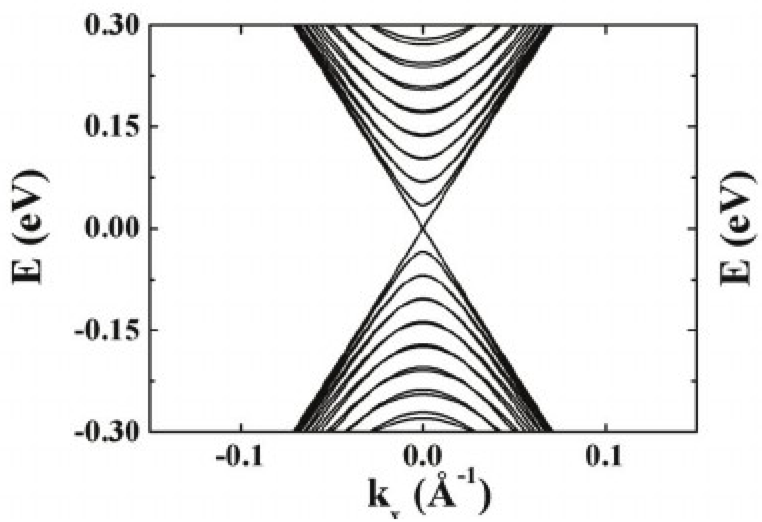
One edge state per sublattice

Armchair geometry



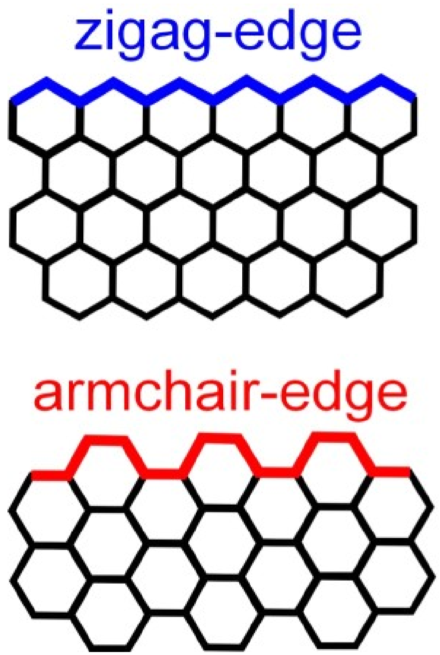
No localized electron states

No edge magnetism

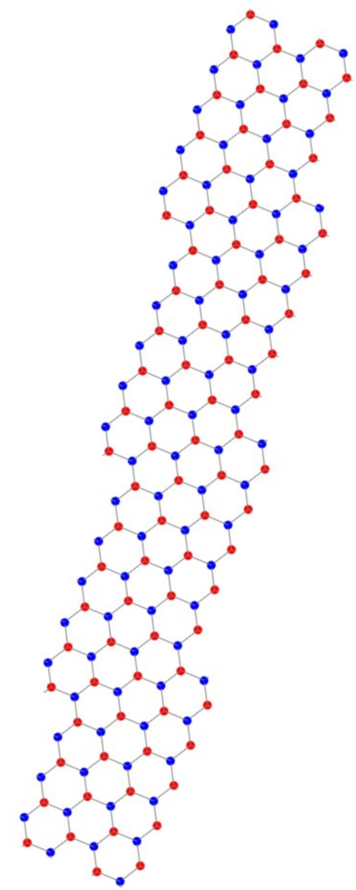


Chiral geometries

Mixtures-- Many possibilities

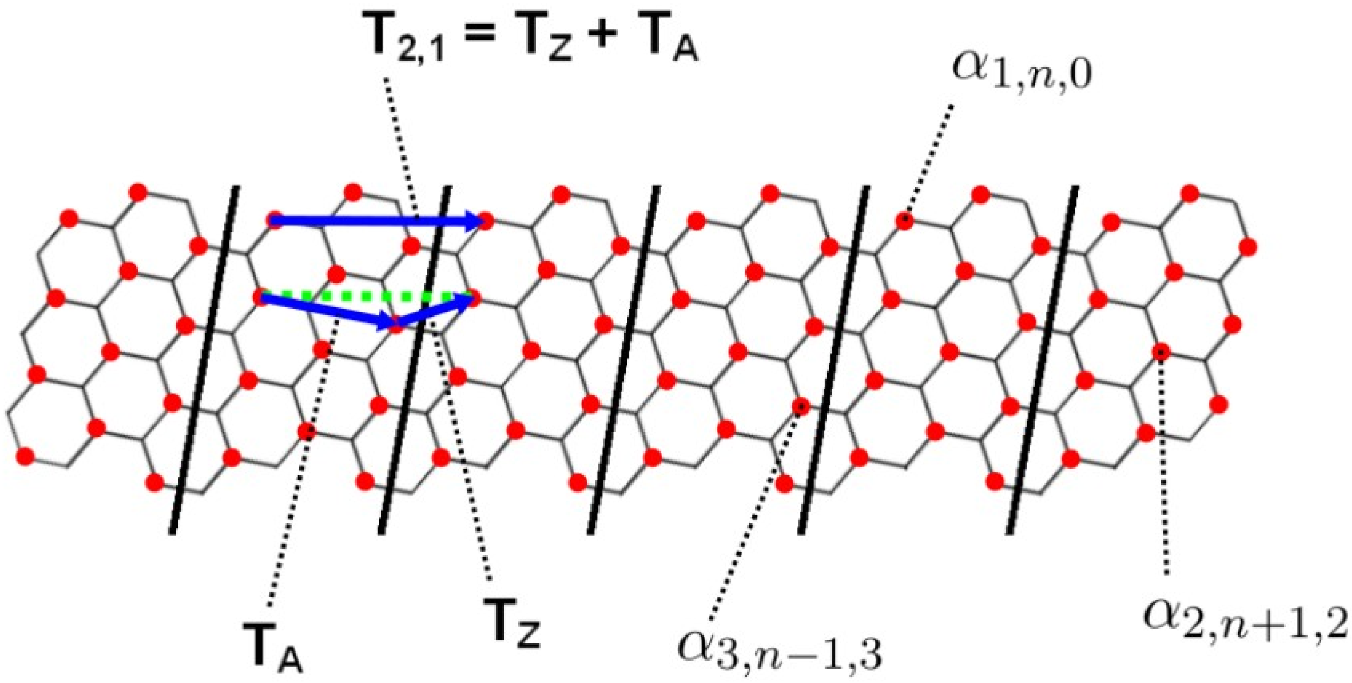


Minimal chiral ribbons



All minimal chiral ribbons support edge states (Ahkmerov et al.)

Chiral geometries



Ribbons characterized by translation vector

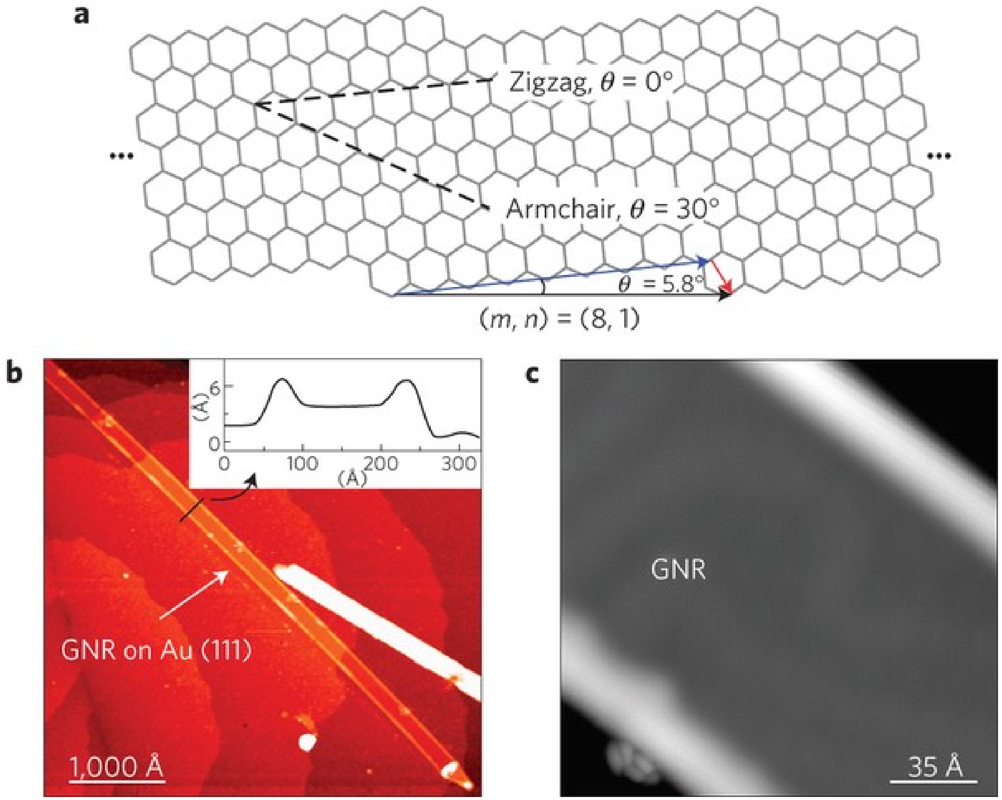
$$\mathbf{T} = (n-m)\mathbf{T}_Z + m\mathbf{T}_A$$

Or by chiral angle

Represents ratio of zigzag/armchair links

Chiral geometries

All minimal chiral ribbons have edge states



Chiral angle describes ratio of zigzag to armchair components

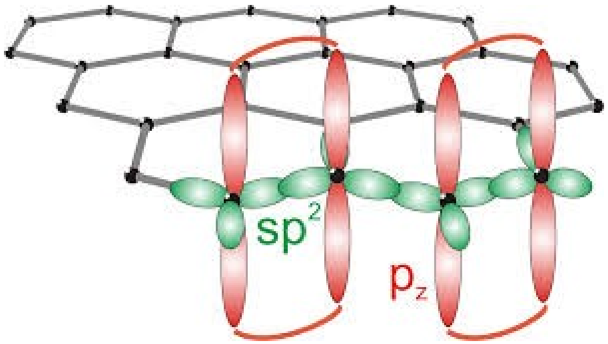
Edge electron density decreases as armchair components are added

$$\rho(\theta) = \frac{2}{3a}\cos(\theta + \pi/3)$$

We prove magnetism for certain chiral ribbons

Bulk electronic structure

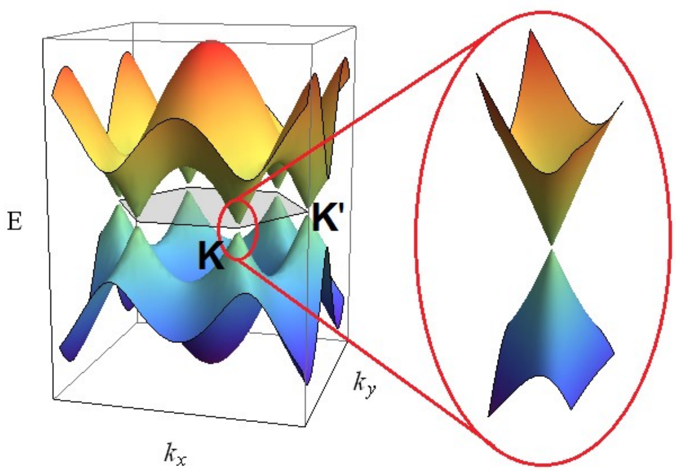
Sp2 hybridization



Local, non-interacting model describes nature

$$H = t \sum_{\langle i,j \rangle} c_i^\dagger c_j + h.c.$$

Relativistic Dirac fermion excitations

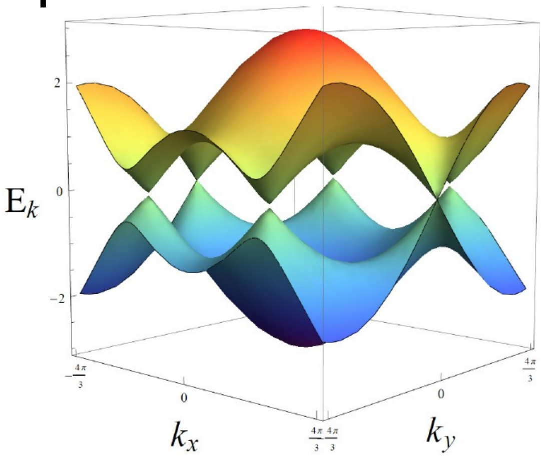


In graphene, interactions are strong

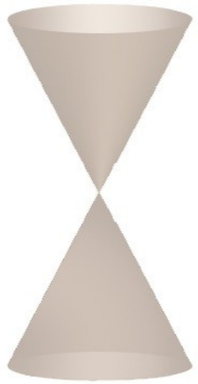
Non-interacting models fail to describe ribbons (gap anomaly)

Bulk electronic structure

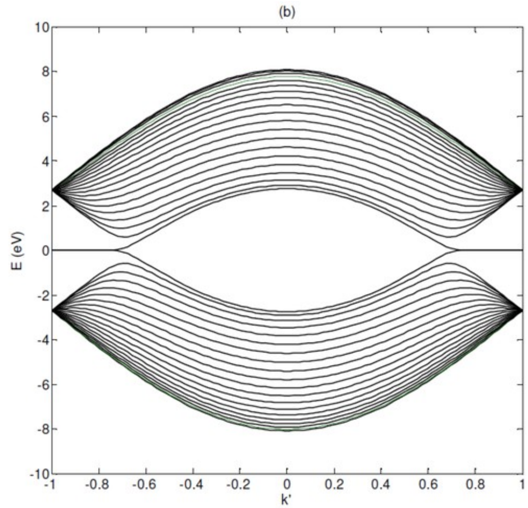
Project 2D Bulk bands onto appropriate reciprocal lattice direction



Dirac excitations

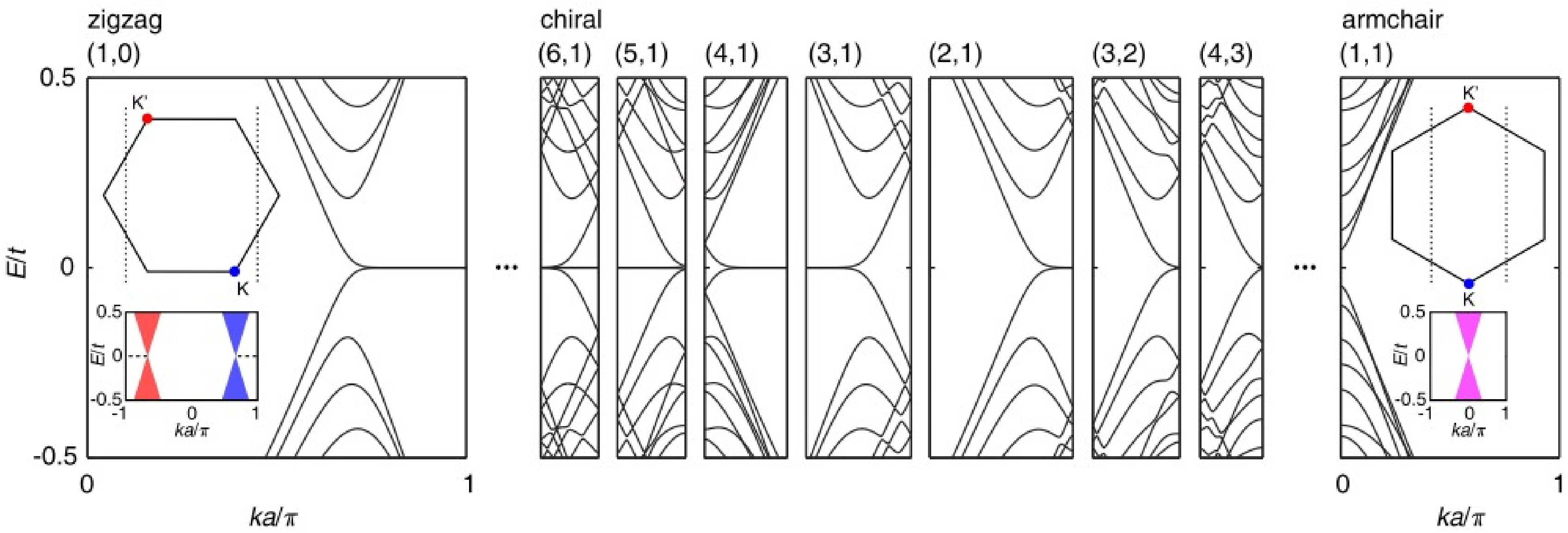


Obtain 1D bulk spectrum of ribbon



Possible existence of flat bands between Dirac points

Ribbon electronic structure



Observations:

1. Confinement shifts Dirac points
2. Flat bands exist between Dirac points
3. Armchair ribbons have overlapping Dirac cones– No edge states
4. Intermediate chiralities all have edge states

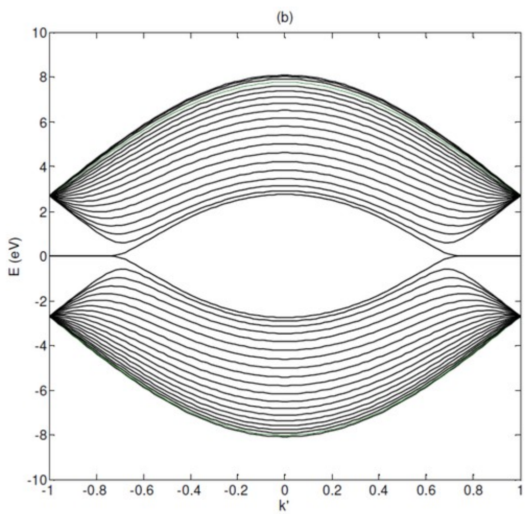
Ribbon edge electronic structure

Flat bands associated with localized electrons

Flat Bands

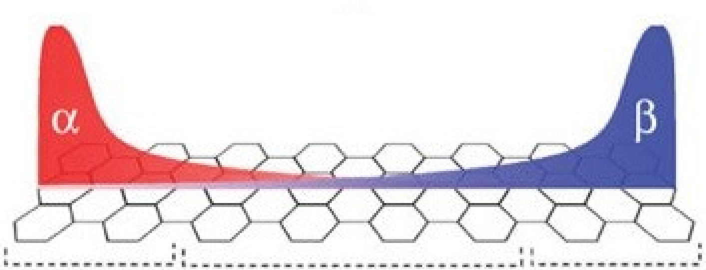
Infinite mass

$$\frac{1}{m_n^*} = \frac{1}{\hbar^2}\left(\frac{\mathrm{d}^2 E}{\mathrm{d}k^2}\right)_{k=k_\mathrm{i}}$$



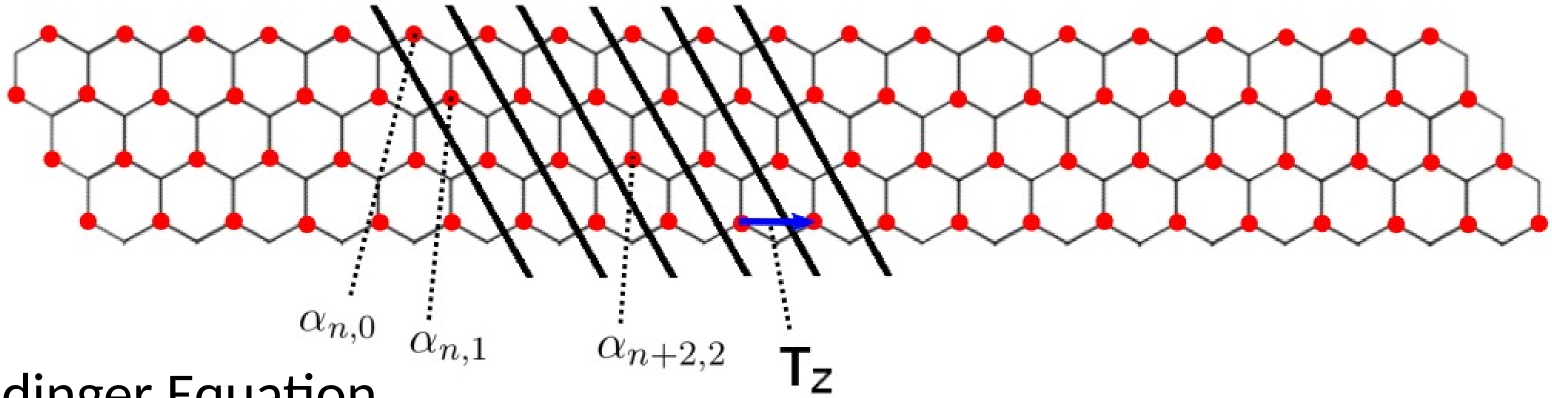
Zigzag ribbon spectrum

Electrons localize at edges:



Edge-localized electrons on zigzag

Unit Cell



Schrodinger Equation

$$E\alpha_{n,m} = t(\beta_{n-1,m} + \beta_{n,m} + \beta_{n,m+1})$$

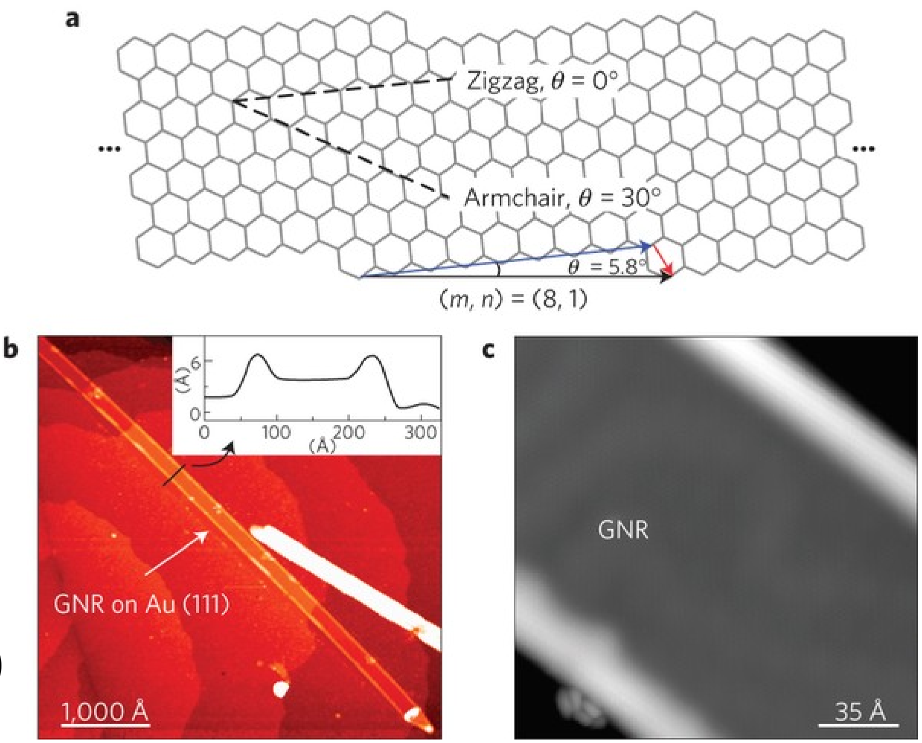
$$E\beta_{n,m} = t(\alpha_{n+1,m} + \alpha_{n,m} + \alpha_{n,m-1})$$

Zero Energy Localization over 1/3 of Brillouin zone

$$|\alpha_m(k)| = \sqrt{\frac{2a}{\Lambda(k)}} e^{-ma/\Lambda(k)} |\alpha_0(k)|$$

<u>Exponential Decay from Edge</u>

Edge-localized electrons on chiral



Density exponentially decaying from edge near zero energy

Density decays with chiral angle (Ahkmerov)

$$\rho(\theta) = \frac{2}{3a}\cos(\theta + \pi/3)$$

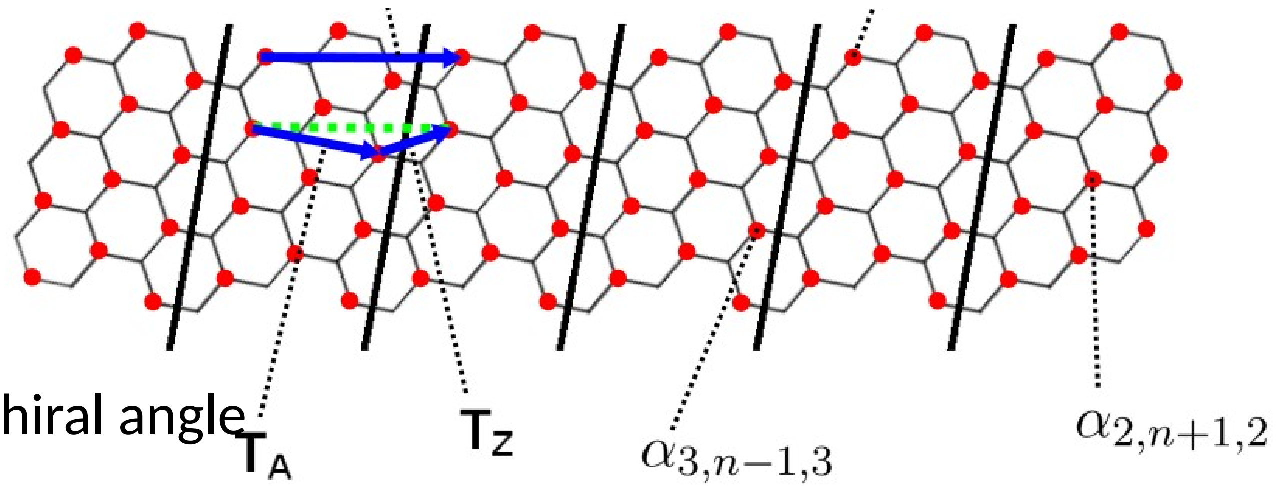
Zero is zigag; 30 degrees is armchair

Edge-localized electrons on chiral

Density exponentially decaying from edge near zero energy (more components)

$$\alpha_{i,n,m}(k) = e^{iknT}\left[g_i(k)\nu_1(k)^m + l_i(k)\nu_2(k)^m\right]$$

(2,1) ribbon



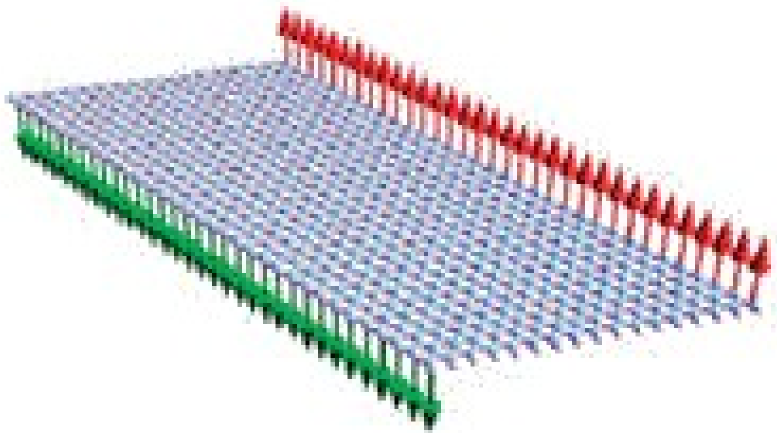
Density decays with chiral angle

$$\rho(\theta) = \frac{2}{3a}\cos(\theta + \pi/3)$$

Zero is zigag; 30 degrees is armchair

Magnetic order on zigzag edges

Small Hubbard interactions induce magnetic order (Karimi/Affleck)



Ferromagnetic along an edge

Antiferromagnetic across the ribbon

Lieb’s theorem:

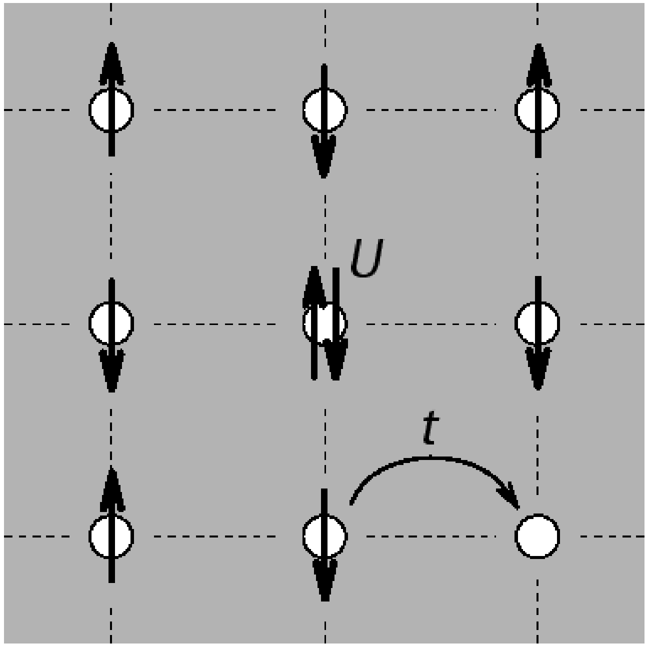
The total magnetization on a bipartite Hubbard model is (N-M)/2

<u>Consistent</u>

Magnetic order on zigzag edges

Karimi-Affleck:
Include local coulomb repulsion
with Hubbard

$$H_U = U \sum_i c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$



Neglect interactions involving bulk
electrons

$$H_U = \sum_{q,m} O^\dagger_m(q) O_m(q)$$

A rigorous proof of edge ferromagnetic order follows

Magnetic order on zigzag edges

Edge-projected Hubbard:

$$H_U = \sum_{q,m} O_m^\dagger(q) O_m(q)$$

What ground states of Hubbard are possible?

$$0 = O_m(q)|\psi\rangle$$
$$\propto \sum_k \alpha_m^*(k)\alpha_m(k+q)\Big[\sum_\sigma e_\sigma^\dagger(k+q)e_\sigma(k) - \delta_{q=0}\Big]|\psi\rangle$$

Answer: Ferromagnetic ones

Magnetic order on zigzag edges

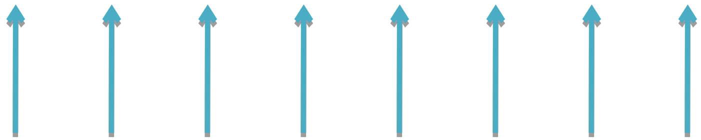
Crux of proof is demonstrating for all k and q,

$$0 = \Big[\sum_{\sigma}\{e_{\sigma}^{\dagger}(k+q)e_{\sigma}(k) + e_{\sigma}^{\dagger}(-k)e_{\sigma}(-k-q)\} - 2\delta_{q=0}\Big]|\psi\rangle$$

Proof has two steps:

1. Show single occupancy at each momentum
2. Show possible states combine symmetrically

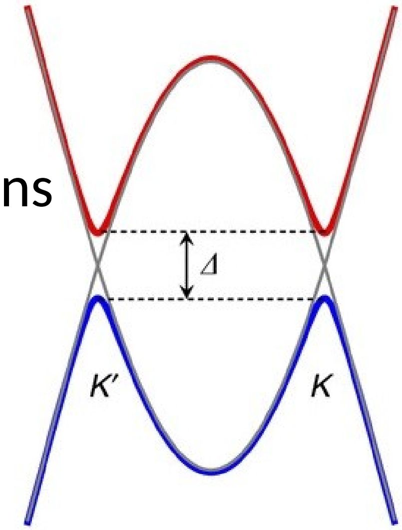
It follows ground state is a symmetric sum of states with one occupation each momentum



i.e. ground state is ferromagnetic along edge

Magnetic order on chiral edges

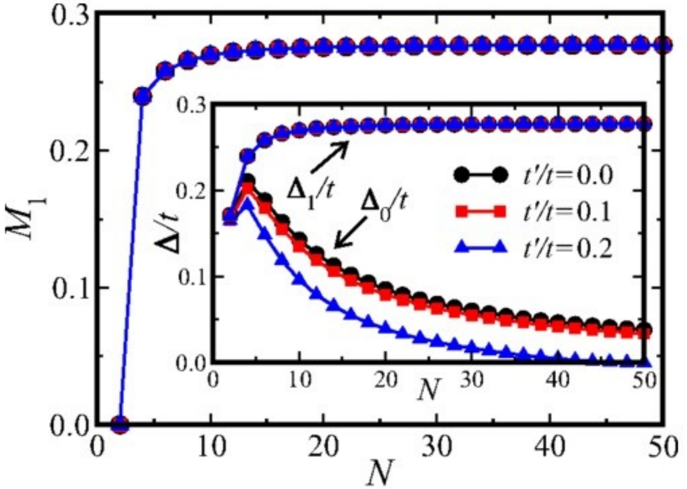
Why is it expected?



1. Explains first principles numerical calculations

Anomalous confinement gap

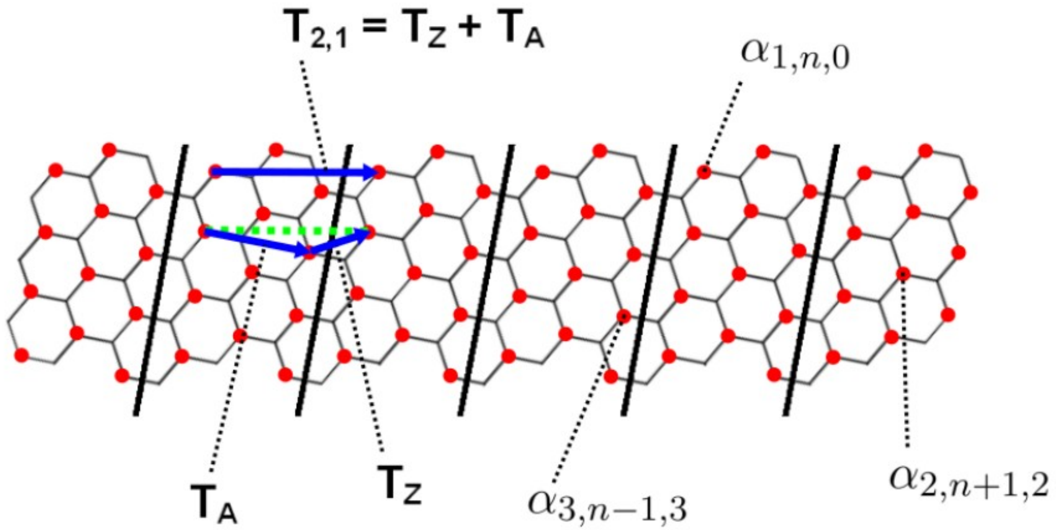
2. Seen in Mean field approximations



We prove in U<<t limit on certain chiral ribbons

Magnetic order on chiral edges

Now we have more distinct sites per unit cell



Wavefunctions have more components

$$\alpha_{i,n,m}(k) = e^{iknT}\left[g_i(k)\nu_1(k)^m + l_i(k)\nu_2(k)^m\right]$$

Magnetic order on chiral edges

We can still edge project Hubbard interaction:

$$H_U = U \sum_i c_{i\uparrow}^\dagger c_{i\uparrow} c_{i\downarrow}^\dagger c_{i\downarrow}$$

$$\Downarrow$$

$$H_U = \sum_{i,m,q} O_{i,m}^\dagger(q) O_{i,m}(q)$$

But proving ferromagnetic ground state from here much tougher

Magnetic order on chiral edges

$$H_U = \sum_{i,m,q} O^\dagger_{i,m}(q) O_{i,m}(q)$$

Crux of proof: Assume psi is a ground state

$$0 = \sum_k \alpha^*_{i,m}(k+q)\alpha_{i,m}(k)\left[\sum_\sigma e^\dagger_\sigma(k+q)e_\sigma(k) - \delta_{q=0}\right]|\psi\rangle$$

Need to separate momentum channels

Difficult with multi-component wavefunctions

$$\alpha_{i,n,m}(k) = e^{iknT}\left[g_i(k)\nu_1(k)^m + l_i(k)\nu_2(k)^m\right]$$

Magnetic order on chiral edges

Set up a matrix structure:

$$\begin{bmatrix} M^i_{0,k_0}(q) & M^i_{0,k_1}(q) & \dots & M^i_{0,k_D}(q) \\ M^i_{1,k_0}(q) & M^i_{1,k_1}(q) & \dots & M^i_{1,k_D}(q) \\ \vdots & \ddots & \ddots & \vdots \\ M^i_{D,k_0}(q) & M^i_{D,k_1}(q) & \dots & M^i_{D,k_D}(q) \end{bmatrix} \begin{bmatrix} |\psi^q_{k_0}\rangle \\ |\psi^q_{k_1}\rangle \\ \vdots \\ |\psi^q_{k_D}\rangle \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Elements:

$$M^i_{m,k}(q) = \alpha^*_{i,m}(k+q)\alpha_{i,m}(k)$$

$$|\psi^q_k\rangle = \Big[\sum_\sigma \{e^\dagger_\sigma(k+q)e_\sigma(k) + e^\dagger_\sigma(-k)e_\sigma(-k-q)\} - 2\delta_{q=0}\Big]|\psi\rangle$$

Must invert matrix and show kets = 0

Magnetic order on chiral edges

We numerically invert matrix on (2,1) and (3,1) ribbons, obtaining

$$0 = \Big[\sum_{\sigma}\{e^{\dagger}_{\sigma}(k+q)e_{\sigma}(k) + e^{\dagger}_{\sigma}(-k)e_{\sigma}(-k-q)\} - 2\delta_{q=0}\Big]|\psi\rangle$$

(Same condition as zigzag case)

The only states satisfying this condition are ferromagnetic .

Conclusion:
Weak hubbard interactions induce ferromagnetic order on (2,1) and (3,1) edges

Scope

Expect a similar ferromagnetism proof on all ribbons having non-degenerate edge modes

Degeneracy poses a problem:

$$\alpha_{i,m} = A\lambda_1^m(k) + B\lambda_2^m(k) + C\lambda_3^m(k) + \ldots$$

Unfixed coefficients in wavefunctions

How to invert matrices to obtain ferromagnetism condition?

$$M^i_{m,k}(q) = \alpha^*_{i,m}(k+q)\alpha_{i,m}(k)$$

No general way

Conclusion

Chiral ribbons

Flat bands

$$H_U = \sum_{q,m} O_m^\dagger(q) O_m(q)$$

Zigzag– Karimi and Affleck, 2010

Small Hubbard interactions

Ferromagnetic order on edges (given non-degenerate edge modes)

$$H_U = \sum_{i,m,q} O_{i,m}^\dagger(q) O_{i,m}(q)$$

Chiral-- Pierce and Affleck, 2016

See UBC thesis “Magnetic Structure of Chiral Graphene Nanoribbons” for details and references

Thanks